Practical – 3 & 4

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Batch : B3

AIM: Perform morphological operations 2. Thresholding 2. Dilation 3.Erosion 4.Opening 5. Closing 6. Gradient

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THEORY:

1. Morphological Operations:

Morphological operations are a set of image processing techniques that are used to analyze and manipulate the shapes and structures within an image. They are based on set theory and mathematical morphology. These operations are particularly useful for tasks such as:

- **Image Segmentation:** Separating objects from the background.
- **Noise Reduction:** Removing unwanted artifacts or noise.
- **Object Recognition:** Identifying specific shapes or patterns.
- **Image Enhancement:** Improving image quality by sharpening edges or removing blur.

Key Concepts in Morphological Operations:

- **Structuring Element (Kernel):** A small matrix or shape used to probe the image. The shape and size of the structuring element determine the effect of the operation.
- **Set Operations:** Morphological operations are often expressed as combinations of set operations like union, intersection, and difference. These operations are applied to the image pixels and the structuring element.

2. Thresholding:

Thresholding is a fundamental image segmentation technique used to separate objects from the background. It involves converting a grayscale image into a binary image by applying a threshold value. Pixels above the threshold are typically assigned a value of 1 (white), while those below are assigned a value of 0 (black).

Different types of thresholding include:

- **Binary Thresholding:** Simple thresholding where pixels are either black or white.
- **Inverse Binary Thresholding:** Inverts the binary thresholding result.
- **Truncated Thresholding:** Pixels above the threshold are set to the threshold value, while those below remain unchanged.
- **To Zero Thresholding:** Pixels above the threshold remain unchanged, while those below are set to 0.
- **Inverse To Zero Thresholding:** Inverts the To Zero thresholding result.
- **Otsu's Thresholding:** Automatically determines the optimal threshold value based on image histogram analysis.

3. Dilation:

Dilation is a morphological operation that expands the boundaries of white regions in a binary image. It works by sliding a structuring element over the image and setting the output pixel to 1 if any part of the structuring element overlaps with a white pixel in the input image.

4. Erosion:

Erosion is the opposite of dilation. It shrinks the boundaries of white regions in a binary image. Erosion is performed by sliding a structuring element over the image and setting the output pixel to 1 only if the entire structuring element is contained within a white region of the input image.

5. Opening:

Opening is a combination of erosion followed by dilation using the same structuring element. It is useful for removing small objects or noise while preserving the shape and size of larger objects.

6. Closing:

Closing is the reverse of opening, involving dilation followed by erosion using the same structuring element. It is helpful for closing small holes or gaps within objects and smoothing object boundaries.

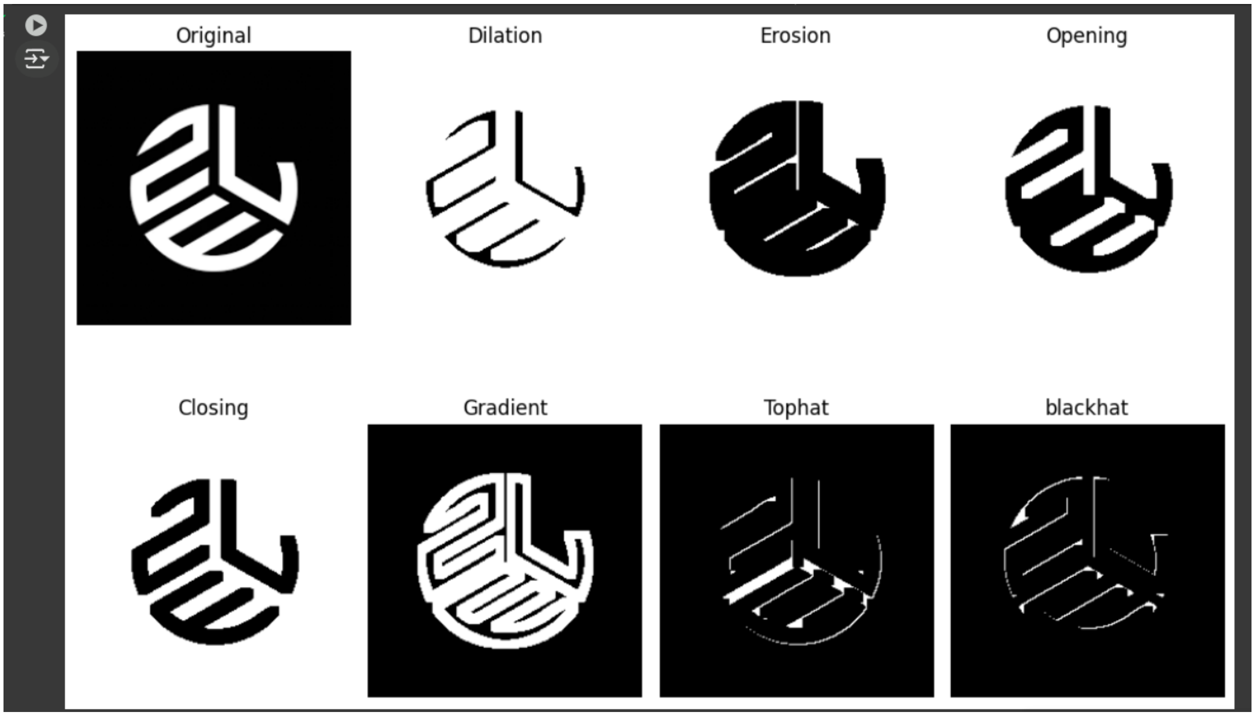
7. Gradient:

The morphological gradient is the difference between the dilation and erosion of an image. It highlights the boundaries or edges of objects in the image.

CONCLUSION:

Morphological operations and thresholding are essential image processing techniques used for various tasks, including image segmentation, noise reduction, object recognition, and image enhancement. They provide powerful tools for analyzing and manipulating the shapes and structures within images.

OUTPUT:


Original
Dilation
Erosion
Opening
Closing
Gradient
Tophat
blackhat